GenAI Hackathon

RAG-Powered Project Intelligence Assistant

Background

Organizations that deliver projects — whether in consulting, technology, engineering, or professional services — accumulate vast institutional knowledge over time. This knowledge lives inside presentation decks, proposal documents, and slide repositories that document hundreds of past and ongoing projects: their problem statements, solution approaches, technologies used, timelines, team compositions, and business outcomes.

For Sales teams, this knowledge is gold — it helps them reference relevant past work during client pitches. For Delivery teams, it reduces re-invention by surfacing what has already been built. Yet in practice, this knowledge remains locked and inaccessible because no one has the time to read 150–200 slides manually every time a question arises.

Problem Statement

The Challenge

You are given a PowerPoint repository of ~150–200 slides covering approximately 100 real-world projects. Each project is described across 1–3 slides containing: project title, problem statement, solution domain, technologies used, team size, duration, and expected outcomes.

Your task is to build a production-grade RAG (Retrieval-Augmented Generation) pipeline that allows a Sales or Delivery professional to query this repository in plain English and receive accurate, grounded, and source-attributed answers.
The system must perform well on a hidden evaluation Q&A set that will be used to benchmark your pipeline's accuracy.

Example queries your system must handle correctly:

- "Show me all projects involving NLP or large language models."
- "Which projects were delivered for the healthcare or pharma domain?"
- "What solution approach was used for supply chain optimization?"

- "List projects where the team size was under 5 people and duration under 3 months."
- "What technologies were used across data engineering projects?"
- "Which project is most similar to building a customer churn prediction model?"

Objective

Build a **RAG-powered Project Intelligence Assistant** that enables Sales and Delivery teams to query a repository of project PowerPoint slides using natural language and receive **accurate, source-backed answers**. The system should extract project knowledge from presentations, retrieve the most relevant projects, and generate grounded responses with slide/project references.

Illustrative Scenario

Build a **RAG-powered Project Intelligence Assistant** that enables Sales and Delivery teams to query a repository of project PowerPoint slides using natural language and receive **accurate, source-backed answers**. The system should extract project knowledge from presentations, retrieve the most relevant projects, and generate grounded responses with slide/project references.

A sales manager is preparing for a client meeting in the healthcare sector and wants to showcase relevant past experience. Instead of manually reviewing hundreds of slides, they ask:

"Show me healthcare projects that used AI or NLP solutions."

The Project Intelligence Assistant:

1. Searches the PowerPoint repository.
2. Retrieves healthcare-related project summaries mentioning AI/NLP.
3. Generates a concise answer listing:
    a. Project names
    b. Business problem solved
    c. Technologies used (e.g., NLP, LLMs, Python)
    d. Outcomes achieved
    e. Source slide numbers

Core Capabilities to Build

- Slide Ingestion & Parsing
- Extract text, structure, and metadata from PowerPoint slides programmatically.
- Handle slides with bullet points, tables, and mixed layouts.

- Chunking Strategy
- Design a chunking approach that preserves project-level context (do not split a single project across unrelated chunks).

- Embedding & Vector Store
- Embed chunks using any model of your choice (OpenAI, HuggingFace, Cohere, etc.).
- Store and query embeddings using a vector database (FAISS, Chroma, Pinecone, Weaviate, etc.).

- Retrieval & Generation
- Retrieve top-k relevant chunks and pass them to an LLM for answer generation.
- Every answer must include a source reference (slide number or project name).

- Query Interface
- Build a minimal UI or CLI interface that accepts a plain-English query and returns a formatted answer.

Bonus Opportunities

- Re-ranking layer (cross-encoder or LLM-based) to improve retrieval precision.
- Confidence scoring — system reports how confident it is in each answer.
- Metadata filtering — filter by domain, technology stack, team size, or duration before retrieval.
- Multimodal parsing — extract and use information from slide images or diagrams.
- Hallucination detection — flag answers where retrieved context does not support the generated response.
- Evaluation dashboard — visualize accuracy scores across the evaluation Q&A set.

Expected Outcome

The **RAG-Powered Project Intelligence Assistant** will enable users to instantly search and query hundreds of project PowerPoint slides using natural language. The system will automatically extract project information, retrieve the most relevant project records, and generate accurate, source-backed answers with slide references. This will reduce manual effort, improve knowledge reuse, accelerate proposal preparation, and help teams quickly identify similar projects, technologies, domains, and solution approaches from the organization's historical project repository.